

Banking Customer Analytics & Dashboard

End-to-end analysis: Python → PostgreSQL → Power BI

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Project Overview

Scope

3,000 retail banking clients —
demographics, accounts,
balances

Pipeline

Clean → store (PostgreSQL) →
visualize (Power BI)

Goal

Produce interactive dashboard for loans & deposits

Objectives

01

Data Quality

Clean, standardize, verify
completeness

02

Feature Eng.

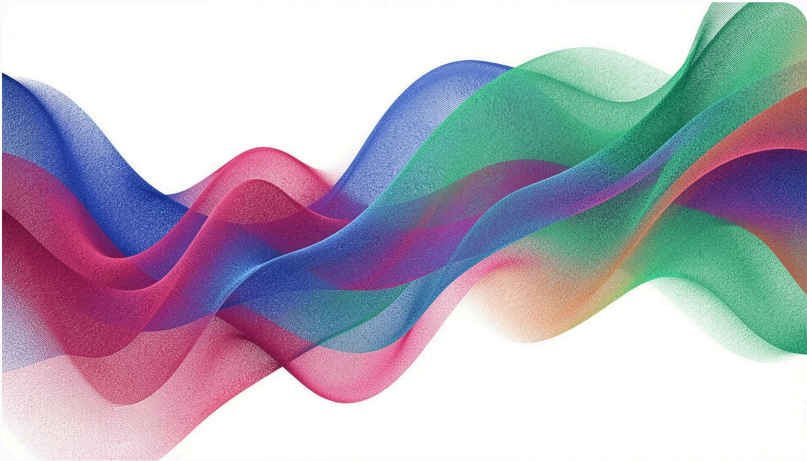
Create income_band: Low / Medium /
High

03

Delivery

Persist to PostgreSQL; build 3-page Power BI report

Tools & Technologies



Data Handling

pandas, NumPy, Jupyter



Database

PostgreSQL, SQLAlchemy, psycopg2



Viz & Dashboard

Matplotlib, Seaborn, Power BI

Dataset at a Glance

Records

3,000 clients

Attributes

25 columns

Completeness

0 missing values — fully complete

Table

banking_data.customer_banking_data (PostgreSQL)

Engineered

income_band (Low / Medium / High)

Methodology

1

Ingest

Load Banking.csv into pandas

2

Clean & Validate

Standardize names; confirm zero nulls

3

Feature Eng.

Quantile binning → income_band

4

Warehouse

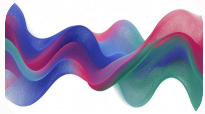
Write cleaned table to PostgreSQL

5

Explore

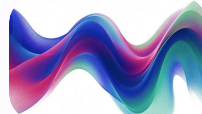
Plots, histograms, heatmap; SQL verification

Key EDA Insights — Demographics



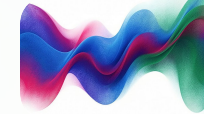
Age

Range 17–85; mean $\approx$ 51
— broad, evenly spread



Gender

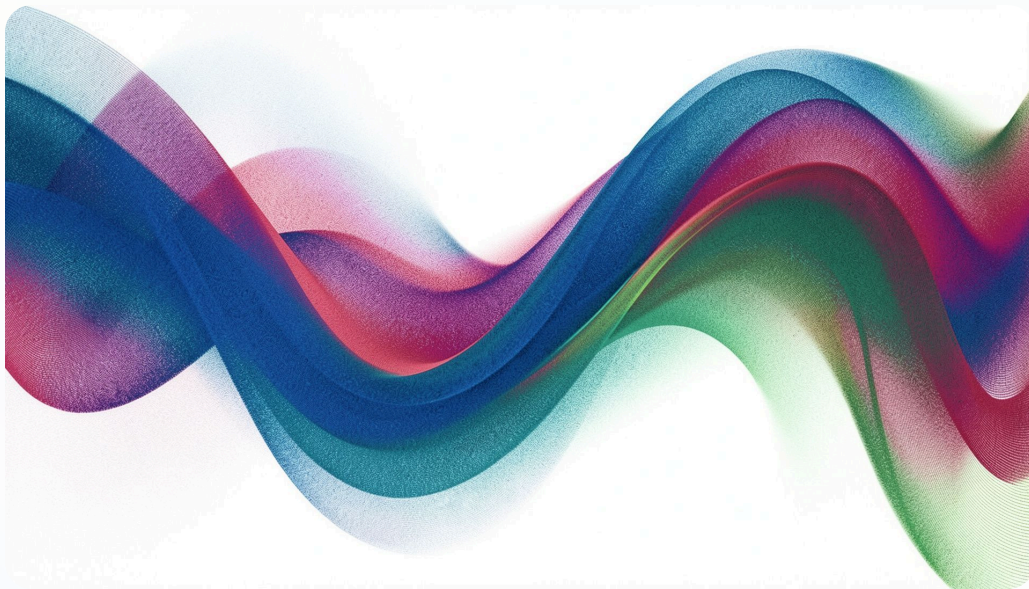
Almost even split: 1,512
vs. 1,488



Nationality

European 1,309 (43.6%);
Asian 754 (25.1%);
American 507; Australian
254; African 176

Key EDA Insights — Financials & Correlations



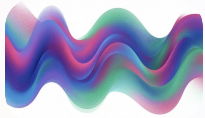
Nine financial vars all right-skewed — long high-value tail



Bank Deposits ↔ Checking (0.84) and Saving (0.75)

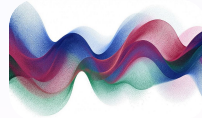
Other pairs: mild-moderate correlation (0.17–0.44)

Power BI Dashboard — Structure



Home

Slicers: Gender, Year of Joining. KPI cards: Total Clients, Total Loan, Total Deposit, Accounts



Loan Analysis

Slicers: Year, Gender, Branch. KPIs & charts: Loan by BR, Occupation, Nationality; Income-band donut



Deposit Analysis

Mirrors Loan page: Total Deposit, Bank Deposit, Saving & Checking amounts; Deposit by BR/Occupation/Nationality

Findings, Implications & Next Steps

Concentration

Branch 3 holds 45% of clients — investigate regional exposure

Loyalty Upsell

44.4% in Jade — target upgrade campaigns

Product Bundling

Deposits, checking, saving move together — promote bundles

High-value Segment

Small tail of high-balance clients — prioritize relationship management

Risk Profile

68.6% at risk levels 1–2 — portfolio leans lower-risk

Future Scope

- Predictive model for default/churn (logistic/tree-based)
- Automate CSV → PostgreSQL pipeline for refresh cadence
- Drill-through client-level detail pages
- Row-level security in Power BI per branch
- Clustering (k-means) for richer segmentation